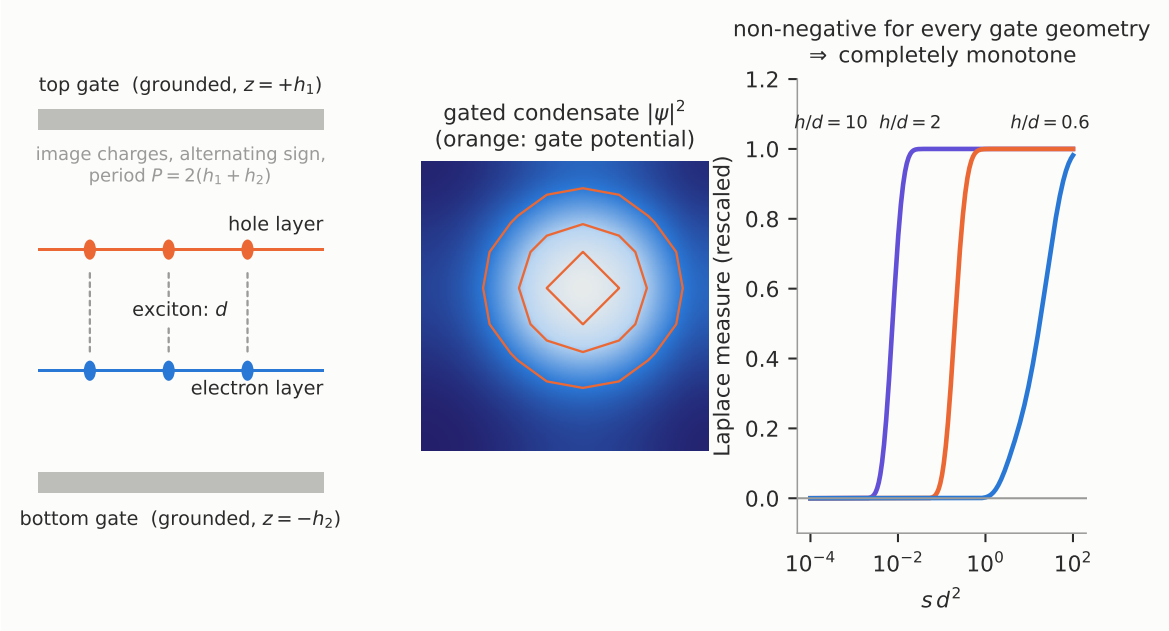


Gated exciton fluids:

a proof that image charges preserve complete monotonicity, a driven-dissipative solver test on rusty-SUNDIALS, and a laboratory protocol

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October 2026

Abstract

A planar universal-optimality theorem for completely monotone interactions (OpenAI, September 2026 [2]; AI-generated, unrefereed) was used in a first note [1] as a dependency check on the exciton–exciton interaction of a 2026 bilayer experiment. That note left open whether the *dual-gated* device kernel, a sum of image charges of alternating sign, is completely monotone in r^2 (numerical evidence only), and whether gates could be engineered to break it. Here we **prove** (Theorem 1, paper and pencil, not in Lean) that for ideal grounded metal gates at arbitrary distances and arbitrary layer positions, the Laplace measure of the kernel in r^2 is a sum of manifestly non-negative terms, $S_m = 2\big(1-\cos(b-a)\big)\big(1+\cos(a+b)\big)$; the gates therefore cannot break complete monotonicity in this model. A by-product is the zero-wavevector coefficient $\tilde{U}(0) = 4\pi d\,(1 - d/D)$ for a gate separation D (a capacitor formula, not new; the mean-field baseline of the protocol). A spectral argument supplied by an independent reviewer extends the theorem to any laterally homogeneous layered dielectric between grounded planes (Section 2). Second, we test the solver stack on the model the experiment's community would use for short-lived excitons or polaritons, a driven-dissipative Gross–Pitaevskii equation with an incoherent reservoir and a gate potential, in a pre-registered design with known answers: the steady states, the 768×768 linearisation (matched to 2.7×10^{-9}), the conservative limit and the IMEX threshold and order passed as registered; the stationarity and pump-budget gates of the gate sweep *failed* as registered at $t = 4000$ (residual up to 3.9×10^{-4}; the horizon was too short for the slow relaxation) and pass in an exploratory continuation to $t = 16000$, one seed per amplitude; the expected shift of the locked emission frequency lay between 0 and V_0, which in this small periodic box is only the area average of the gate. We report what failed or had to be changed (a registered cost was misjudged by two orders of magnitude; three amendments were written before the numbers they affect, a fourth while the last run was in progress, and one registered comparison was replaced by a better-posed one). Third, we give a laboratory protocol, and say which of its items can fail and which are controls. We do not use Hugging Face materials data or neural-network surrogates, do not compare with data, and make no claim about exciton Bose–Einstein condensation. Written by an AI system; not refereed.

1 Why this note, and what it is not

The request that prompted this note asked for a driven-dissipative exciton fluid on rusty-SUNDIALS with materials data from Hugging Face. The first note [1] did something else (conservative gradient flows, a lattice theorem). We say so because the two are easily confused: that note did *not* contain a driven-dissipative model, an ARKode-type integrator, a gate, a defect, or any materials input. This one adds the first three in a minimal form (Section 3) and tests the ARKode-style module that exists in the library; it does not add the last (Section 5).

What is established here. (i) Theorem 1 below, a short exact argument (checked numerically on 300 random geometries). (ii) Registered numerical gates on a standard open-dissipative model: which passed, which failed, in Section 4. **What is not.** No quantum phase transition is located; no experiment is reproduced; the model parameters are dimensionless and not those of any device; the protocol of Section 6 is a proposal that nobody has run.

2 Gated kernels are completely monotone

Setting. Grounded perfect-conductor planes at $z = +h_1$ and $z = -h_2$; a homogeneous dielectric between them (a uniform ε is a prefactor, an anisotropic one rescales z by a constant); an electron layer at z_e and a hole layer at $z_h = z_e + d$. The exciton–exciton kernel is $U = V_{ee} + V_{hh} - 2V_{eh}$, V the Green function of the box. With $P = 2(h_1 + h_2)$ a unit charge at z_0 produces at (z, r)

$$V(z, z_0; r) = \sum_{n\in\mathbb{Z}} \Big[\big(r^2 + (z - z_0 - Pn)^2\big)^{-1/2} - \big(r^2 + (z - 2h_1 + z_0 - Pn)^2\big)^{-1/2} \Big].$$

Theorem 1. *U is completely monotone as a function of $t = r^2$ on $(0, \infty)$: $U = \int_0^\infty e^{-st}\mu(s)\,ds$ with $\mu \geq 0$.*

Proof. Write each term as $(t + a^2)^{-1/2} = \pi^{-1/2}\int_0^\infty e^{-s(t+a^2)}s^{-1/2}ds$. Then $\mu(s) = (\pi s)^{-1/2}S(s)$ where $S(s)$ is the sum over images of e^{-sa^2} with the signs of $V_{ee} + V_{hh} - 2V_{eh}$. Poisson summation, $\sum_n e^{-s(c+Pn)^2} = (\pi/s)^{1/2}P^{-1}\sum_{m\in\mathbb{Z}} e^{-\pi^2m^2/(P^2s)}\cos(2\pi mc/P)$, gives $S(s) = (\pi/s)^{1/2}P^{-1}\sum_m e^{-\pi^2m^2/(P^2s)}S_m$ with, for $\alpha = 2\pi m/P$, $a = \alpha(z_e - h_1)$, $b = \alpha(z_h - h_1)$ (so $\alpha d = b - a$),

$$S_m = 2 - \cos 2a - \cos 2b - 2\cos(b - a) + 2\cos(a + b) = 2\big(1 - \cos(b - a)\big)\big(1 + \cos(a + b)\big) \geq 0,$$

using $\cos 2a + \cos 2b = 2\cos(a + b)\cos(a - b)$; and $S_0 = 0$ by neutrality. Hence $S(s) \geq 0$ termwise. □

Consequences, and limits of the claim. The single-gate and gate-free limits are pointwise limits of completely monotone functions and so are completely monotone. Within *this* model the gates cannot be engineered to violate the hypothesis of the lattice theorem; if Kagome or stripe ground states are observed in a gated device, then, *conditionally on the unrefereed lattice theorem*, they come from something the model leaves out: lateral inhomogeneity, finite screening of real gates, disorder, or quantum fluctuations.

Remark (layered dielectrics; due to the independent review of this note, not proved here in detail). For a laterally homogeneous layered dielectric $\varepsilon(z)$ between the grounded planes, expanding the Green function in the Dirichlet Sturm–Liouville modes $-(\varepsilon\phi')' = \lambda\varepsilon\phi$ gives $U(r) = 2\sum_m \big(\phi_m(z_e) - \phi_m(z_h)\big)^2 K_0(\sqrt{\lambda_m}\,r)$, a positive combination of $K_0(\kappa r) = \frac{1}{2}\int_0^\infty e^{-u-\kappa^2r^2/4u}\,du/u$, each completely monotone in r^2. So dielectric image charges of weight

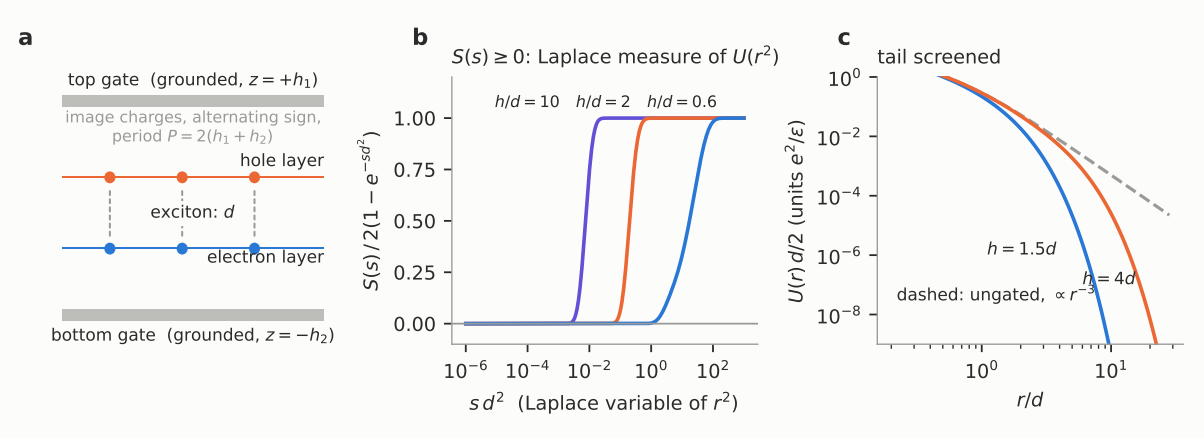


Figure 1: (a) The device and its image series. (b) $S(s)/2(1 - e^{-sd^2})$ from the exact Poisson form for three symmetric gate distances: non-negative everywhere, as Theorem 1 states. (c) The kernel with and without gates: the gates remove the dipolar r^{-3} tail (exponential screening at $r \gg h$) without leaving the completely monotone class. All three panels are generated from the formulas of the text by `phase3_figures.py`.

κ^n do not break the property either; the reviewer checked the expansion numerically on one four-layer stack. We have not repeated that check and have not written the full argument. The same sum-of-squares structure is behind $S_m = 2(\sin a - \sin b)^2$. The lattice theorem itself is imported and was not refereed. A Lean formalisation needs the Poisson summation of Mathlib's Jacobi theta; we did not do it. The zero-wavevector coefficient follows from the same series (and from Gauss's law for three capacitors in series), for a uniform ε in units $e^2/\varepsilon = 1$:

$$\tilde{U}(0) = \int U\, d^2r = 4\pi d \left(1 - \frac{d}{D}\right), \qquad D = h_1 + h_2. \tag{1}$$

We checked it against the numerical integral of the image series (agreement to 3×10^{-3} relative at $D = 3d$ and $D = 20d$, limited by the integration cutoff). It is a capacitor formula, textbook. For piecewise ε the factor $1 - d/D$ becomes $1 - (d/\varepsilon_m)/\sum_i t_i/\varepsilon_i$ (Gauss's law with layers t_i of permittivity ε_i between the gates, ε_m between the exciton layers), which changes the size of the D-dependence for hBN-encapsulated stacks. It is the *mean-field* coefficient: valid at densities where correlations are weak (Section 6, L1).

3 A driven-dissipative model with a gate, tested on rusty-SUNDIALS

Model. Dimensionless units $\hbar = m = 1$,

$$i\,\partial_t \psi = \left[-\tfrac{1}{2}\nabla^2 + V(\mathbf{x}) + g|\psi|^2 + g_R n_R + \tfrac{i}{2}\big(R n_R - \gamma\big)\right]\psi, \tag{2}$$

$$\partial_t n_R = P(\mathbf{x}) - (\gamma_R + R|\psi|^2) n_R, \tag{3}$$

the standard open-dissipative (Wouters–Carusotto [4]) form with $g = 0.04$, $g_R = 1$, $\gamma = 0.1$, $\gamma_R = 200$, $R = 1$ (stiff ratio $\gamma_R/\gamma = 2000$), uniform pump P and gate $V = V_0 e^{-d^2/2\sigma^2}$. Above threshold $P_{th} = \gamma\gamma_R/R = 20$ the uniform state is $n = P/\gamma - \gamma_R/R$, $n_R = \gamma/R$, $\omega = gn + g_R n_R + V$ (we use $P = 21$: $n = 10$, $n_R = 0.1$, $\omega = 0.5$). The model is generic; it is the form one would write for short-lived excitons or polaritons, but with reservoir parameters chosen for solver stiffness (a reservoir 2000 times faster than the condensate, $g_R/g = 25$), not measured ones; real reservoirs

are typically slower than the condensate. It is *not* the model of the equilibrium experiment of Qi *et al.* [5], whose dipolar excitons live microseconds.

Design. Pre-registered in `docs/designs/EXCITON_FLUID_PHASE3_PREREG.md` (commit `335caa5`, before any run of the Rust program), independent Python reference first (`phase3_reference.py`). Gates: D-a steady states above and below threshold; D-b the 768×768 linearisation of the solver against the reference 3×3 per-k spectrum; D-c conservative limit (norm and energy); D-d gate sweep ($V_0 \in \{0, 0.25, 0.5, 1\}$): pump budget $\sum P = \sum(\gamma_R n_R + \gamma|\psi|^2)$, single locked frequency, and the hypothesis $0 < \omega(V_0) - \omega(0) < V_0$; D-e the ARKode-style IMEX module, whose implicit stage is a fixed-point iteration and so must fail when $h\gamma_R a_{ii} \geq 1$ (bracketed by five step sizes only); D-f its temporal order; D-g a planted bug. A deliberately *wrong* prediction is excluded by construction: the intuitive local-density guess that the gate leaves the density pinned at $P/\gamma - \gamma_R/R$ is incompatible with a single locked frequency (the gate shifts the local frequency, so currents must flow), and was not registered.

Amendments and disclosures. A1–A3 were written before the numbers they affect; A4 (D-g was evaluated on three of ten seeds, from the per-seed log lines, because the planted-bug runs take about 25 minutes each) was recorded while D-g was running and committed with the results. A1: the IMEX module's fixed-point test squares the tolerance (effective absolute tolerance $\mathrm{rtol}\,(\mathrm{atol} + \mathrm{rtol}|z|)$), so 10^{-12} cannot converge in floating point; set to 10^{-6}. A2: integrating D-a/D-d/D-g in the laboratory frame needed more than 23 minutes for one run (the phase rotates at $\omega = 0.5$ and BDF must resolve it for $t = 4000$); the flow is integrated in the frame rotating at the registered value of ω with all gate quantities evaluated gauge-covariantly. A3: even then the pinned CVODE refactors a dense matrix every few steps (43 ms per step at 768 unknowns, hours per run, about 70 runs), so **D-a, D-d, D-g were run on an 8×8 grid with $\sigma = 1.5$** (same σ/L), not 16×16; D-b, D-c, D-e, D-f ran on 16×16 as registered. *The 8×8, $\sigma = 1.5$ problem is not equivalent to the registered 16×16, $\sigma = 3$ one* (Section 4): the consequence is that the gate-sweep results describe a very small box. Further disclosures: (i) D-b was registered as a comparison of the two *sorted* eigenvalue lists; as implemented it gives a maximum difference 18.2 because sorting by (real, imaginary) is unstable for nearly degenerate pairs, and the reported 2.7×10^{-9} is from a nearest-neighbour matching introduced in the summariser without an amendment, which should have been one; the Jacobian itself was reconstructed independently by the reviewer and agrees. (ii) Exploratory runs launched to test the horizon were first made with a stale binary that ignored the overrides (results identical to $t = 4000$); they are kept under `results/phase3/exploratory_invalid/` and were not used. (iii) The first, laboratory-frame attempt and the stopped 16×16 rotating-frame attempt are kept as logs under `results/phase3/`.

gate	grid	outcome (numbers from `summary.json`)	registered result
D-a	8×8	above threshold, 10 seeds started at the closed form plus 10^{-3} noise (a stability check, not a basin test): relative n, n_R and absolute ω deviations 1.2×10^{-12}, 5.9×10^{-14}, 4.8×10^{-13}; below: $\max \lvert\psi\rvert \leq 1.6 \times 10^{-11}$	pass
D-b	16×16	eigenvalues of the solver's 768×768 finite-difference Jacobian matched one-to-one to the reference to 2.7×10^{-9}; largest real part 1.4×10^{-10} (the Goldstone mode). *The registered sorted comparison gives* 18.2*; see Section 3*	pass with a replaced criterion
D-c	16×16	norm drift 1.6×10^{-11}, energy drift 4.1×10^{-10} over $t = 20$ (6464 steps)	pass
D-d (i),(ii)	8×8	pump budget up to 5.0×10^{-7} (gate 10^{-8}), stationarity residual up to 3.9×10^{-4} (gate 10^{-7}); even $V_0 = 0$ gives 1.6×10^{-7}	**fail**
D-d (iii)	8×8	$\omega(V_0) - \omega(0) = 0.0488,\ 0.0876,\ 0.1422$ for $V_0 = 0.25, 0.5, 1$: between 0 and V_0, increasing (medians at $t = 4000$); in this box this is the area average of the gate, see text	pass (hypothesis; uninformative)
D-e	16×16	one step succeeds at $h = 0.005, 0.01, 0.0125$ and returns an error at $0.02, 0.05$ (threshold 0.01707); no silent non-finite result; trajectory to $t = 20$ within 7.3×10^{-6} of CVODE	pass
D-f	16×16	error ratios 4.20 and 4.07 on halving h (second order)	pass
D-g	8×8	planted bug: $n = 220.0$ instead of 10; budget violated by ≥ 0.52 (3 seeds, amendment A4)	pass

Table 1: Gates as registered. Every number is generated from the run records by `phase3_paper_numbers.py`.

4 Results of the registered gates

The registered gate-sweep gates failed. At $t = 4000$ the stationarity residual of the gated states is up to 3.9×10^{-4} and the pump budget is violated by up to 5.0×10^{-7}, against thresholds 10^{-7} and 10^{-8}; the ungated state also exceeds its threshold (1.6×10^{-7}). The horizon was set from the decay rate of the ungated Bogoliubov modes (2.4×10^{-3}) applied to a 10^{-3} perturbation; the gated states start $O(0.1)$ from the uniform state and the slowest linear decay rates about them are 1.4–2.1×10^{-3} (reviewer's linearisation), so $e^{-1.4 \times 10^{-3} \cdot 4000} \approx 3 \times 10^{-3}$ of an $O(0.1)$ amplitude is left. The ungated failure (1.6×10^{-7}) was predictable from the registered numbers. All ten seeds end on the same state to all printed digits, and the residual does not depend on the integrator tolerance (rtol $= 10^{-8}$ to 10^{-12} give 7.6×10^{-5}), so it is not integration error. *Exploratory* continuations (not gates, one seed each) to $t = 16000$ give residual 1.0×10^{-8} and budget violation 1.1×10^{-13} for $V_0 = 0.5$, which would pass; the other amplitudes give $V_0 = 0$: residual 3.3×10^{-10}, budget 0.0×10^{0}; $V_0 = 0.25$: residual 6.0×10^{-8}, budget 2.4×10^{-14}; $V_0 = 1$: residual 8.7×10^{-9}, budget 6.9×10^{-13} (all within the registered thresholds).

What the gate sweep shows, and does not show. In the small box the gate depletes the condensate at its centre (from 10 to 7.07, 4.58, 1.49 for $V_0 = 0.25, 0.5, 1$, while the corner rises to 11.08, 11.92, 13.06), and currents flow. The single emission frequency moves by 0.0488, 0.0876, 0.1422. *That shift is a box-size artefact and carries little information.* An independent re-solution of the same stationary states by the reviewer (own Newton solver; it reproduces the shifts to

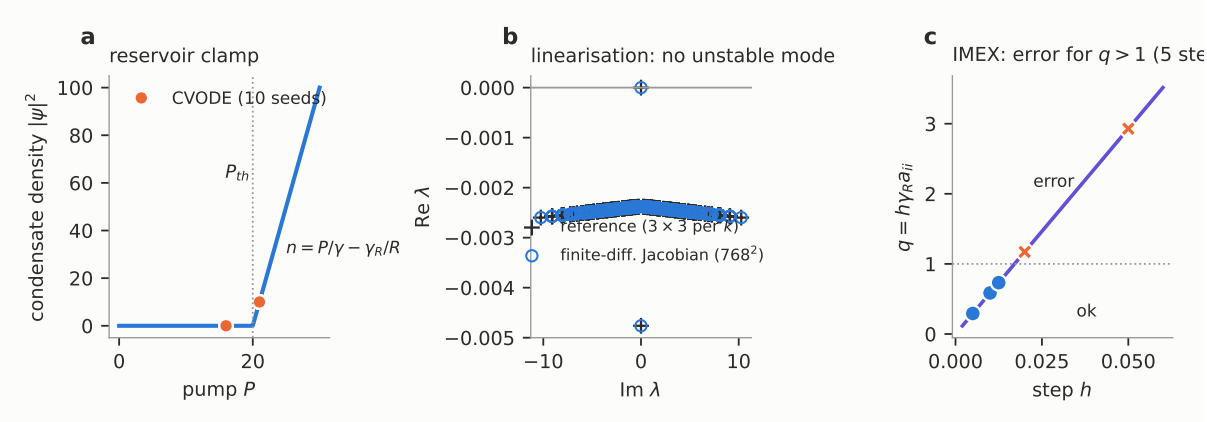


Figure 2: Solver tests. (a) Steady-state density against pump: closed form (line) and ten CVODE runs per pump (dots; they coincide to 1.2×10^{-12}). (b) The eigenvalues of a central-difference Jacobian of the solver's right-hand side (circles) lie on the reference spectrum (crosses); the isolated points are the $k = 0$ phase (Goldstone) and amplitude modes; the fast reservoir mode near $-\gamma_R$ is outside the plotted range. (c) The IMEX module's fixed-point stage: contraction factor $q = h\gamma_R a_{ii}$; it returns an error for $q = 1.17$ and 2.93 and succeeds for $q \leq 0.73$ (the threshold $q = 1$ is bracketed, not located).

five digits) at fixed gate width $\sigma = 1.5$ gives, for $V_0 = 1$, 0.142 ($L = 8$), 0.024 ($L = 16$), 0.0055 ($L = 32$): $\Delta\omega$ falls like L^{-2} and at small V_0 equals the area average of the gate, $V_0\, 2\pi\sigma^2/L^2$. "Between 0 and V_0" is therefore implied by any bounded non-negative potential averaged over a periodic box, and hypothesis (iii) of D-d, though it passed, does not test the model. Nor is the 8×8, $\sigma = 1.5$ problem equivalent to the registered 16×16, $\sigma = 3$ one at "the same σ/L": the reviewer's solution for the registered geometry gives shifts $0.046, 0.077, 0.107$ and central densities $5.8, 2.4, 0.10$ for $V_0 = 0.25, 0.5, 1$, against $0.049, 0.088, 0.142$ and $7.1, 4.6, 1.5$ here, and its slowest mode at $V_0 = 1$ is about 3×10^{-4}, so the registered horizon would have failed there far worse. The linearisation about the numerical states has no unstable mode within the (relaxation-limited) accuracy 9.2×10^{-6}. No finite-size study of our own was made, and the dependence on the gate width and on the pump was not scanned.

The ARKode-style module. It does what its design says: a fixed-point implicit stage that converges iff $h\gamma_R a_{ii} < 1$, so for the stiff reservoir of this model ($\gamma_R = 200$) the step is limited to 0.017 while CVODE's BDF takes the steps it chooses; with h below the threshold it is second order and agrees with CVODE to 7.3×10^{-6}. It is not an adaptive stiff integrator and the library's default `ImexSolver` should not be used for stiff problems without the replacement of the implicit stage.

Cost. The pinned CVODE factorises a dense matrix every few steps (43 ms per step at 768 unknowns, 1.4 s per $t = 20$ at 192 unknowns); this, not the physics, set the grid of D-d. A matrix-free Krylov implicit solver (the spectral Laplacian makes the Jacobian dense, so banded storage does not apply), or an exponential or split treatment of the linear kinetic term, is the obvious requirement for a 64×64 grid and was not available in the pinned tree.

5 What was not done

Materials data and surrogates (Hugging Face, GNNs). Not used. The gate is a Gaussian of chosen amplitude, the parameters are dimensionless. Predicting exciton binding energies

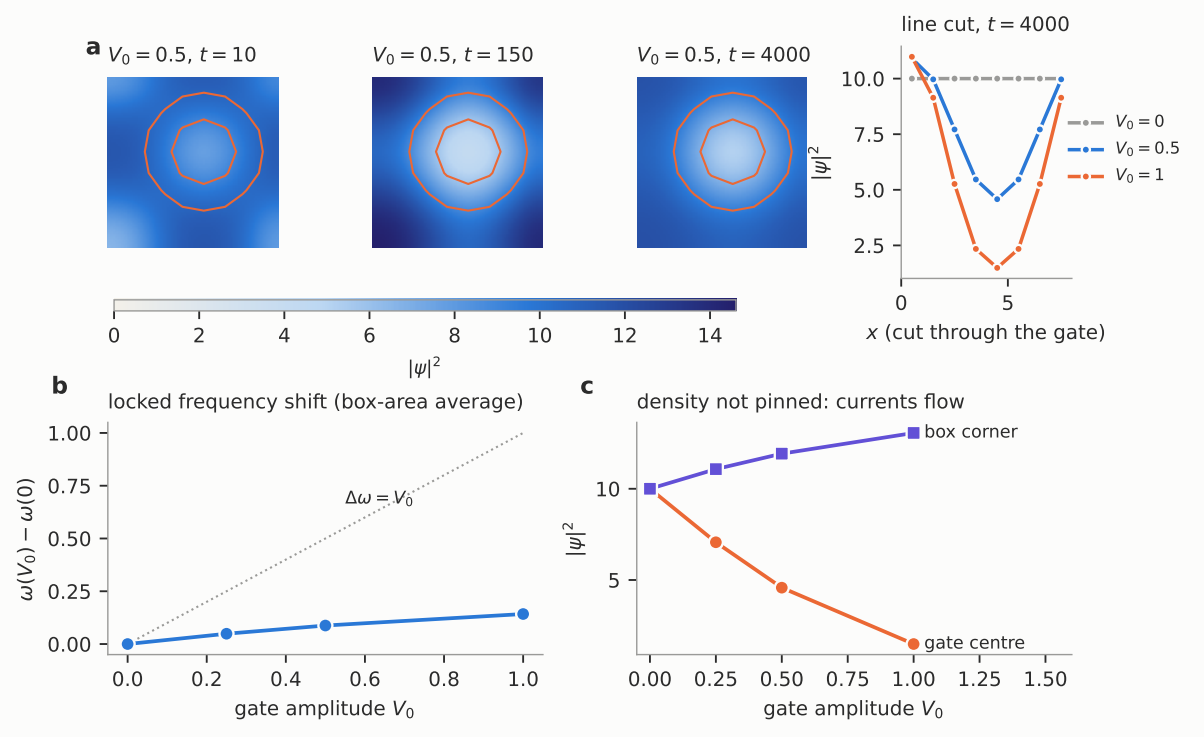


Figure 3: A gated driven-dissipative condensate on the 8×8 grid ($\sigma = 1.5$), uniform pump $P = 21$. (a) Density from the uniform start with noise at three times (orange: contours of the gate; fields are bicubic interpolations of the 8×8 values) and a line cut through the gate at $t = 4000$ for three amplitudes. (b) The locked emission frequency shifts with the gate amplitude by less than V_0; in this small box that is the area average of the gate (text). (c) Density at the gate centre falls and at the box corner rises: the density is *not* pinned at its ungated value, because the single locked frequency requires currents (largest current density 7.6×10^{-3}–1.3×10^{-2} for $V_0 = 0.25$–0.5).

from a graph neural network trained on bulk or monolayer data for a twisted bilayer with defects would require a validity check that has not been made (the training distributions of MACE/ALIGNN-type models are not heterostructures with interlayer excitons); the output of such a model would be an input parameter with an uncontrolled error, not a result.

The Wigner transition. The classical statement of the lattice theorem is a zero-temperature, zero-fluctuation statement; where the quantum fluid freezes needs quantum Monte Carlo (the literature on two-dimensional dipolar bosons has such calculations; we did not read them in this session and quote no number). What can be computed with the present tools is a harmonic-phonon (Lindemann) estimate for the gated kernel from the Hessian code of the first note: a crude order-of-magnitude, to be labelled as such and compared with the QMC literature before use. Not done.

Frustration and edges. The first note found frustration energies of 7.9×10^{-7}–8.6×10^{-3} above the lattice energy on incommensurate tori. Open boundaries, domain walls and edge currents are not in that note and not here. A design (disks of N particles, boundary shapes, count of 5–7 disclinations against the registered lattice energy) is a straightforward extension of the first note's N-particle flows; it is the cheapest next numerical study and has not been started.

6 Laboratory protocol

The aim is to turn the statements that survived into measurements that can fail. Each item gives the quantity, the prediction, the pass/fail rule and the confounders. None has been performed; the model parameters of Section 3 must be *measured* on the device (items L1–L3) before the later items are meaningful.

Platform and regime. A dual-gated transition-metal-dichalcogenide bilayer or heterobilayer with hBN spacers, graphite (or metal) top and bottom gates, a device series with gate separations D spanning at least $5d$ to $50d$ (d the electron–hole layer separation), one-gate and two-gate versions. L2–L3 need a regime in which the dipolar excitons are cold and degenerate or crystalline; L5 needs a driven regime with excitons or polaritons of lifetime short compared with the observation time (a microcavity, or intralayer excitons under continuous pumping), and L4 only makes sense in that second regime. These are different samples or conditions; the model of Section 3 does not apply to long-lived interlayer excitons.

Common controls. Gates grounded and floating; a device without the second gate; a detuned pump; a temperature series; at least three devices per geometry; blind analysis (fits fixed before the data of the discriminating series are read); raw data and analysis code deposited with the preregistration of the series. Density is calibrated independently of the quantity tested (capacitance or absorption), not from the photoluminescence blueshift itself.

What can fail. Of the six items, only **L1** is a prediction that depends on the mathematics of this note and can be falsified by a measurement; L2 is a calibration, L3 and L4 are consistency checks that cannot falsify the theorem, L5 is a textbook control for the driven model, and L6 has no prediction yet.

	Measurement	**Expected outcome**	**Pass / fail, confounders**
L1	Interaction strength: PL peak shift versus calibrated exciton density n, device series in D, low T.	Weak-correlation (mean-field) regime: $dE/dn = \tilde{U}(0) = (4\pi e^2 d/\varepsilon)(1 - d/D)$ (Eq. 1), increasing with D; the single-gate device approaches the ungated $4\pi d$ (null control). In the strongly correlated, lattice-like regime the slope is the classical lattice value, which for the exact gated kernel is 0–0.6 of $\tilde{U}(0)$ over $nd^2 = 0.003$–0.3, $D/d = 3$–50 (independent reviewer's mode sum), *not* equal to it.	Compare the trend with D and the null control, not the absolute value: fail if the slope is independent of D within 10% over $D/d \in [3, 20]$ in the weak-correlation regime, or the trend is opposite. For hBN stacks use the piecewise-ε factor of Section 2. Confounders: correlation corrections that depend on D and n, density calibration, heating.
L2	Phase diagram: structure factor $S(\mathbf{k})$ (momentum-resolved emission or interferometry) versus n and T at fixed D.	No number is predicted. The classical lattice has a hexagonal Bragg ring at $k_0 = 4\pi/(\sqrt{3}a)$ with $n = 2/(\sqrt{3}a^2)$.	Calibration: it says where L1 (weak versus strong correlation) and L3 apply. A ring at the wrong k_0 for the measured n is a calibration failure, not a result.

	Measurement	Expected outcome	Pass / fail, confounders
L3	Lattice symmetry versus D in the solid regime: classify the peaks of $S(\mathbf{k})$.	Triangular at every D in the classical limit, within the model.	*Consistency check, not a test of Theorem 1*: a triangular lattice is the ground state of essentially every repulsive monotone 2D interaction, so hexagonal order does not single out complete monotonicity. A non-hexagonal pattern points to lateral inhomogeneity, finite gate screening, disorder or quantum fluctuations; the experiment cannot tell the theorem from these.
L4	Gate-voltage spectroscopy: nanofabricated finger gate, DC bias, pump above threshold, spatially resolved spectroscopy (driven regime only).	A single locked emission frequency. The size of the shift is set by the gate area relative to the condensate area (at small V_0, the area average of the gate), a property of the geometry, not a test of the model.	Only the absence of locking (two lines at all biases) would count against the driven model. Confounders: Stark-shift calibration, heating, multimode condensation.
L5	Threshold and clamp: condensate-peak and reservoir (high-energy tail) intensities versus pump P (driven regime).	Textbook open-dissipative behaviour: linear growth above P_{th} with slope $1/\gamma$ (an absolute slope, needing absolute calibration of P and n), reservoir density clamped at γ/R.	A *control* for the applicability of the driven model, not a prediction of this note. Fail if the reservoir keeps growing linearly above P_{th}. Confounders: detector saturation, multimode condensation, thermal tail.
L6	Edge/frustration (exploratory): boundary-shaped trap arrays with N excitons.	No numerical prediction yet (Section 5).	Do not run before the numerical design exists.

Decision rule. L2 comes first (it fixes the regime); L1 is then the only falsifiable test and needs only a device series in D; L5 tells whether the driven model applies to a given driven sample at all. A failure of L1's trend with D would invalidate the use of ideal-metal image kernels, with their completely monotone structure, for that platform.

7 Limitations

- Theorem 1 is proved for ideal grounded metals and a homogeneous dielectric; the extension to layered dielectrics rests on a reviewer's argument, not on a full proof here; neither is in Lean; the lattice theorem they feed is unrefereed and conditional in the first note's Lean files.
- The numerical model is generic and dimensionless, with reservoir parameters chosen for stiffness; one parameter set; D-a, D-d, D-g on an 8×8 grid in which the gate shift is dominated by the box size (Section 4); no finite-size or resolution scan beyond the reviewer's;

the solver (dense CVODE, ≤ 768 unknowns) is not claimed to be the best of its kind.
- The ARKode-style module tested is a fixed-step ARS222 with a fixed-point implicit stage on a linear diagonal stiff term; it is not SUNDIALS' ARKode, and D-e is a unit test of a known contraction condition, not evidence about nonlinear stiff stages.
- The D-b registered comparison was replaced without an amendment (Section 3); the D-d (i,ii) failure was predictable from the registered numbers (the ungated noise alone gives about 7×10^{-8} times an $O(1)$ factor at $t = 4000$).
- No experimental data; no Hugging Face data; no contact with any experimental group; the citations to Qi *et al.* and to the OpenAI release are as in the first note and were not re-verified here; written by an AI system and reviewed only by another instance of the same model family and by the author's brief check.

Reproducibility

Pre-registration, amendments, run records and summary: `docs/designs/EXCITON_FLUID_PHASE3_PREREG.md`, `exploration/exciton/results/phase3/`; instrument `exploration/exciton/rust/qf-exciton-p1/src/bin/dd.rs` (rusty-SUNDIALS tag v11.6.0; `-features m8` for the 8×8 runs); reference `phase3_reference.py`; gate evaluator `phase3_summarise.py`; the numbers of this note `phase3_paper_numbers.py`; figures `phase3_figures.py`; proof check `gate_image_cm_*.py`. First note: concept DOI 10.5281/zenodo.23289026.

References

[1] X. Callens, *Testing a planar lattice-optimality theorem against an exciton-fluid interaction*, Zenodo, doi:10.5281/zenodo.23289027 (2026).

[2] OpenAI, *Universal optimality of the triangular lattice*, https://github.com/openai/math, family 090 (2026).

[3] H. Cohn and A. Kumar, J. Amer. Math. Soc. **20**, 99 (2007).

[4] M. Wouters and I. Carusotto, Phys. Rev. Lett. **99**, 140402 (2007).

[5] R. Qi *et al.*, Nature **654**, 629 (2026).

[6] A. C. Hindmarsh *et al.*, ACM Trans. Math. Softw. **31**, 363 (2005).